

Specializations of epithelial cells.

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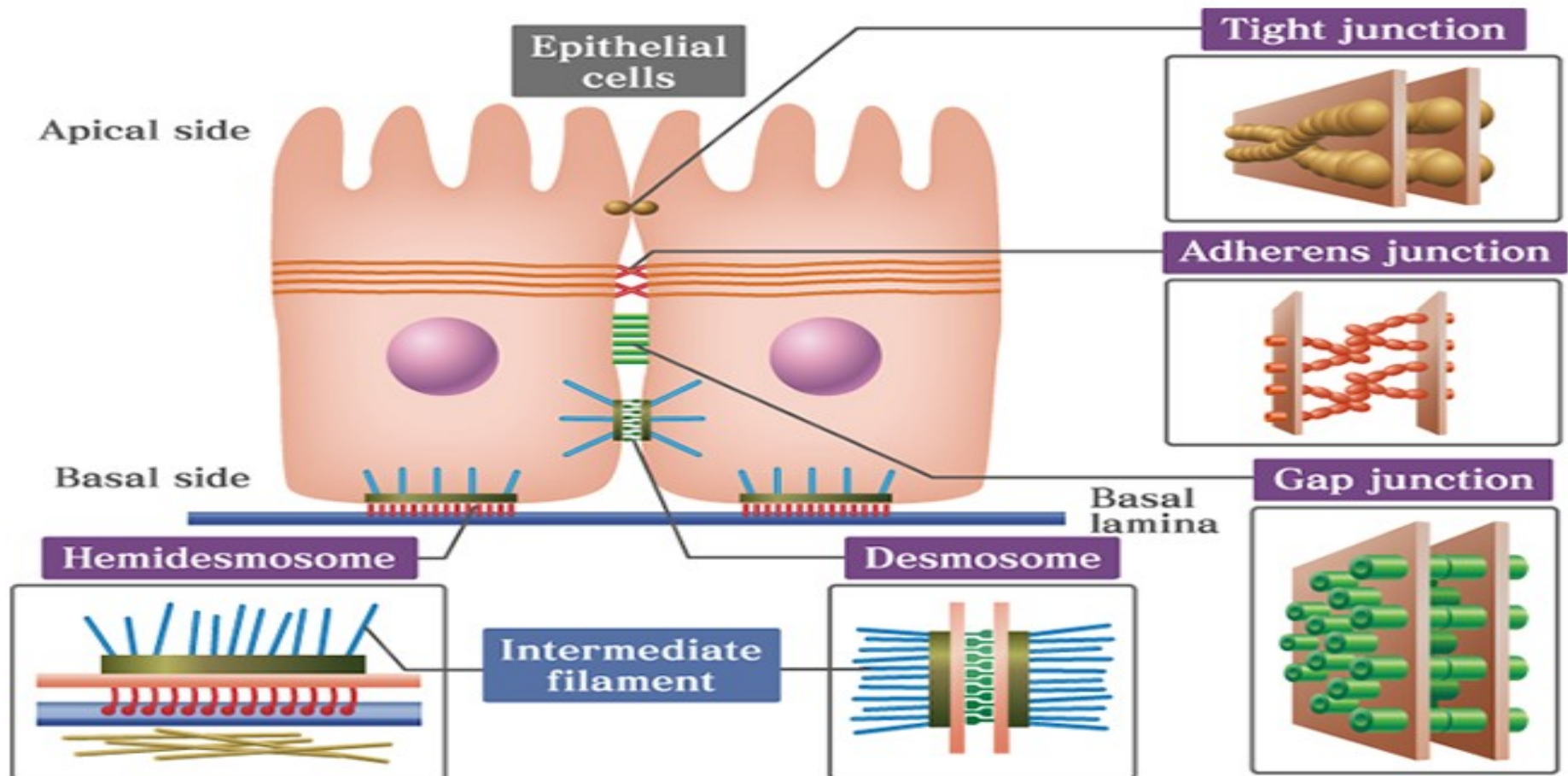
:Objectives



- Epithelial tissue cells are only type of cell that have specialized surfaces that each of them has a special distinguished structure that is only found in the surfaces of epithelial tissues.
- Student must understand different types of apical surface specializations such as, cilia, flagella, stereocilia, and microvilli.
- Student must understand all type of lateral surface specialization of epithelial cells.
- Student must understand type of basal surface specialization of epithelial cells.
- Student must understand structure and function and location of each of the specialization on each surfaces.

.Specializations of Epithelial Cell Surfaces

- Each epithelial cell has 3 surfaces (an apical, lateral and a basal surface).



.A- specialization of the apical surface

- The apical surface of many types of epithelial cells contain specialized structures that increase the surface area for absorption or move substances.

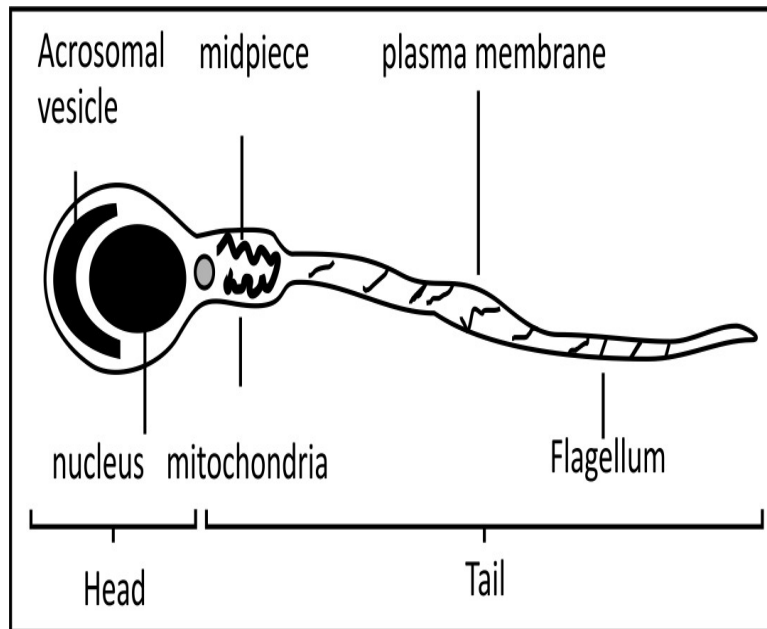


Fig. : Structure of sperm

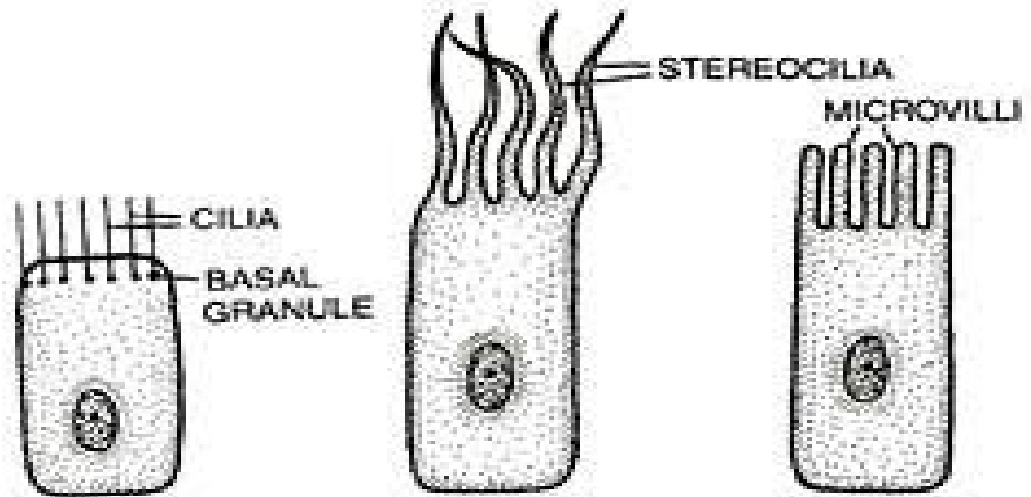
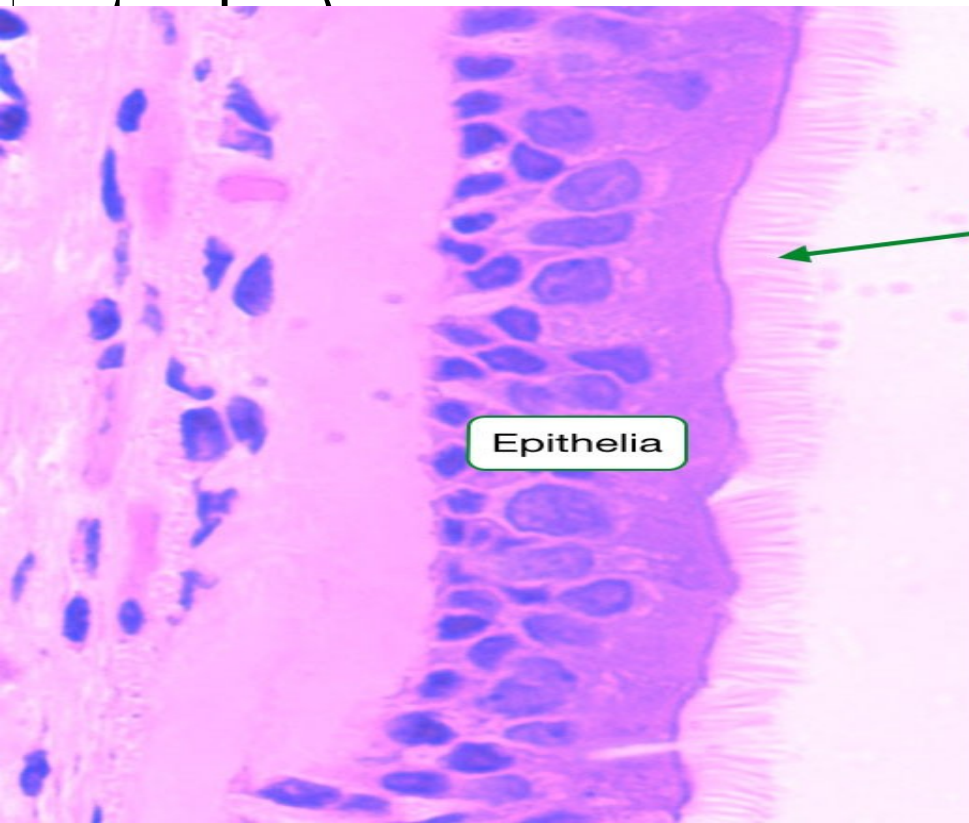


Fig. 7.2. Epithelial cells showing projections in the form of cilia, stereocilia and microvilli.

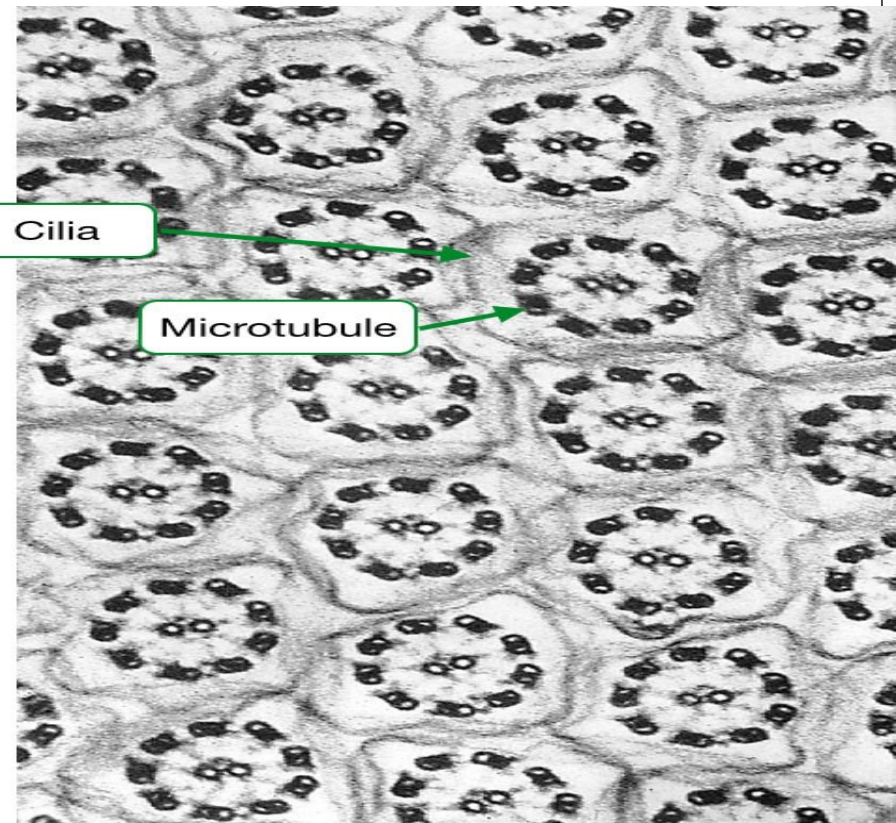
A- specialization of the apical surface.

:1- Cilia

Are cylindrical motile surface covered by plasma membrane, They contain a central pair of isolated microtubules surrounded by nine pairs, cilia are beating in waves to moving the surface coat mucus & trapped materials, (as in ciliated epithelium of



Epithelia

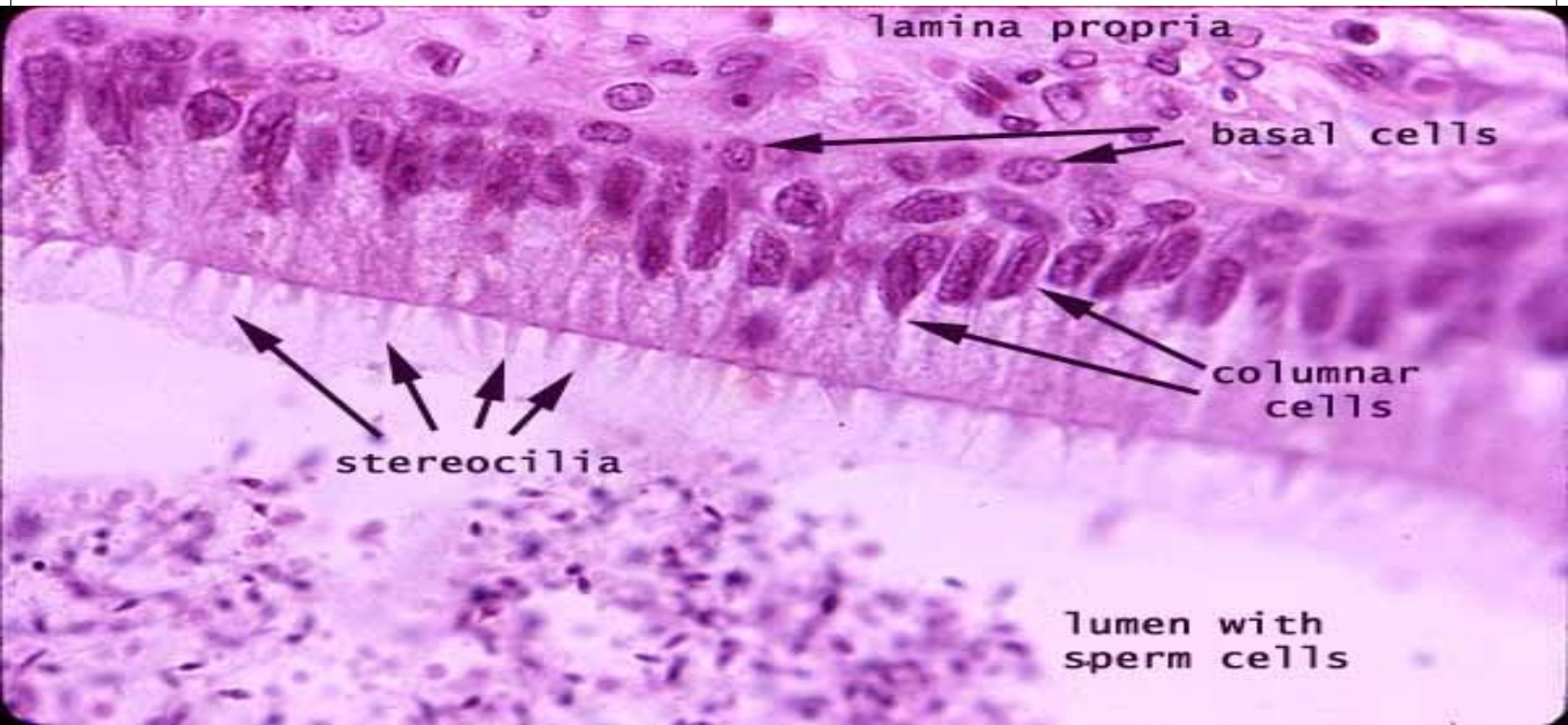


Cilia

Microtubule

:Stereocilia -2

- are long microvilli , nonmotile, may show branching, it increase the surface area (epididymis), and help perception of stimuli (internal ear)



3- Flagella :

- A flagellum is similar to cilium but is longer & there is usually only one or two per cell . flagella are concerned with movement as in the spermatozoa (the male gamete).

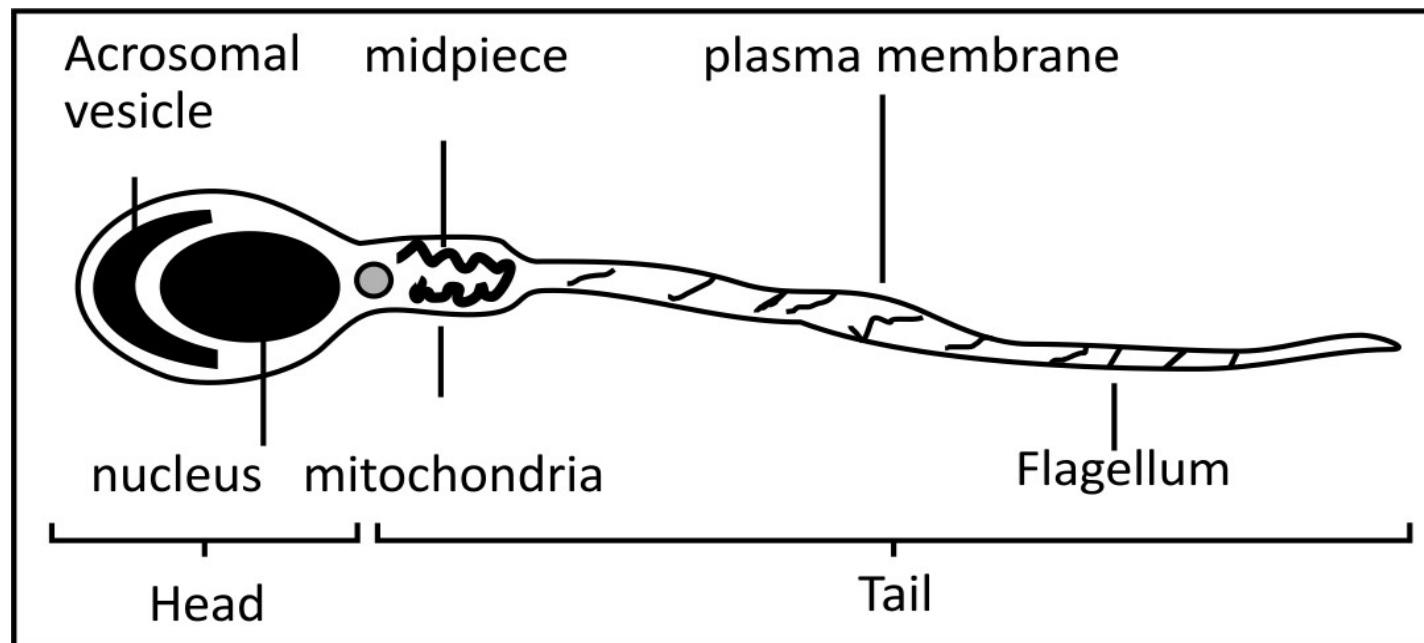
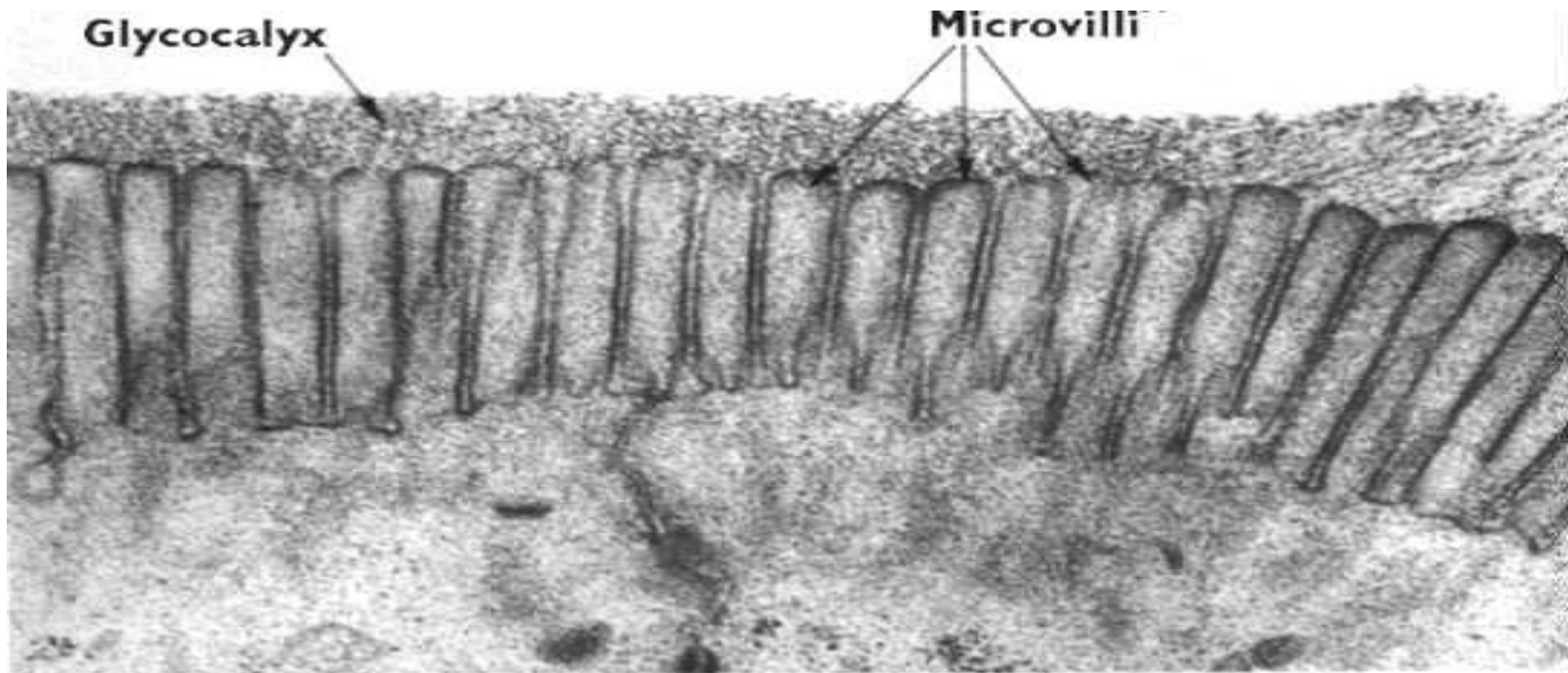


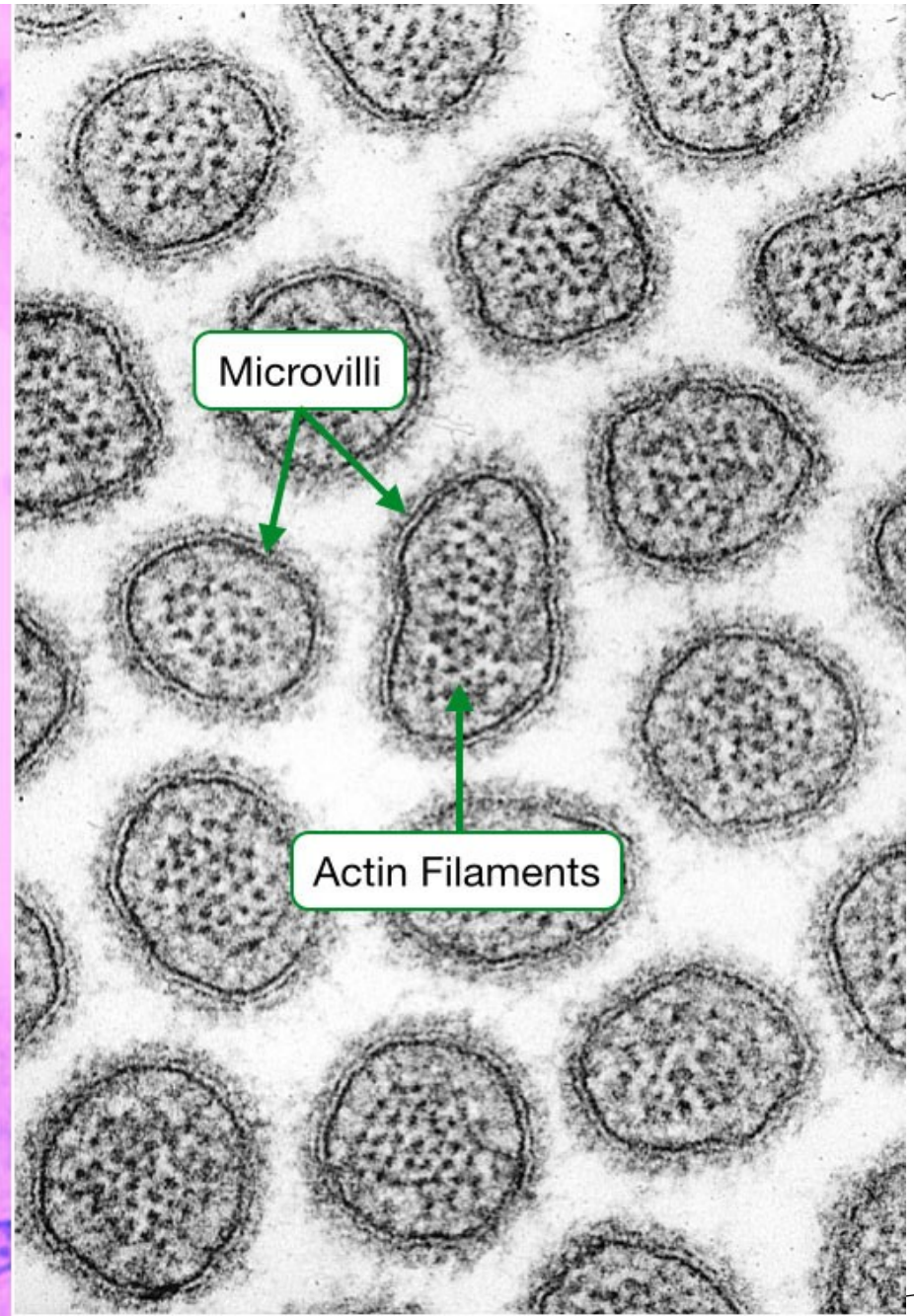
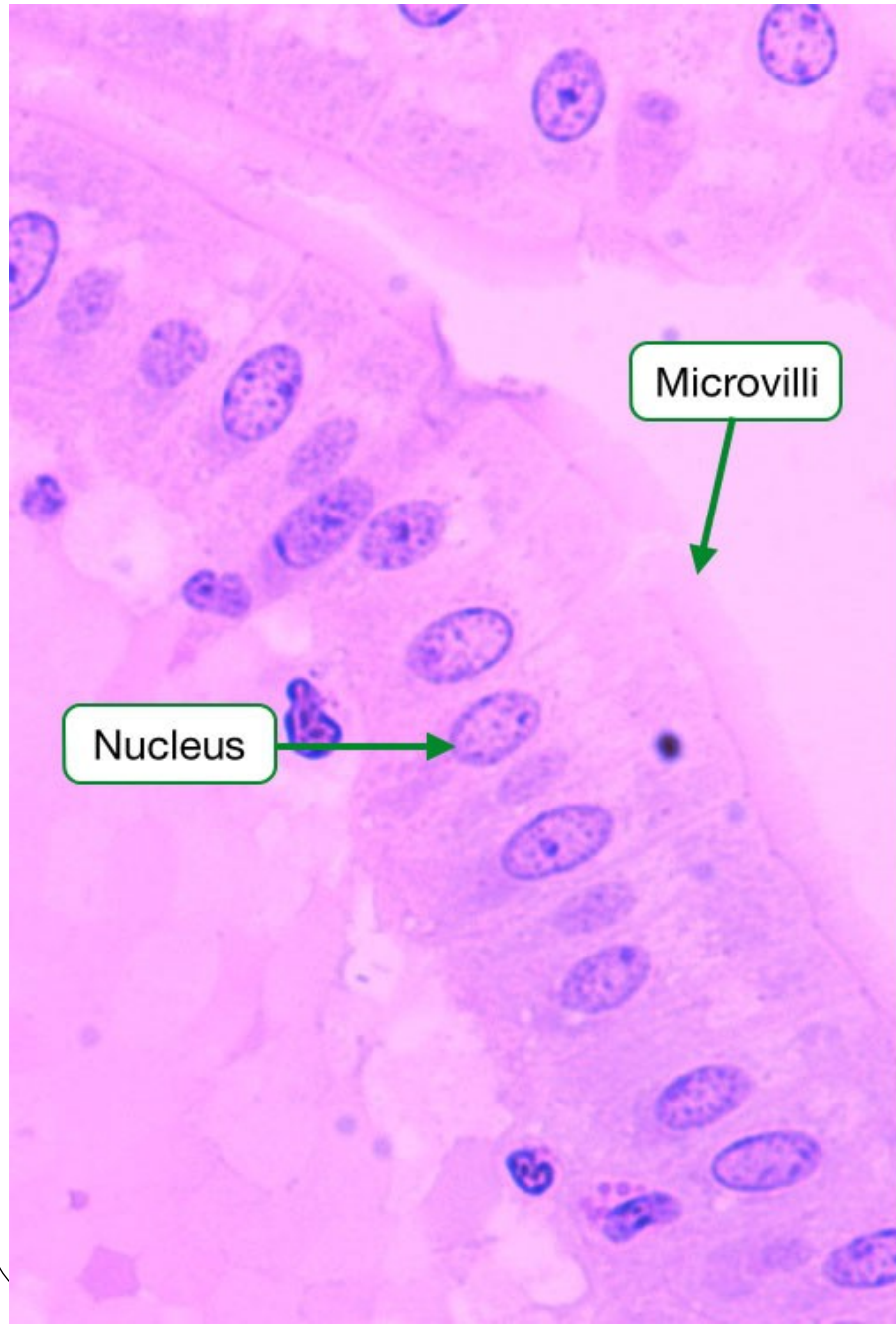
Fig. : Structure of sperm

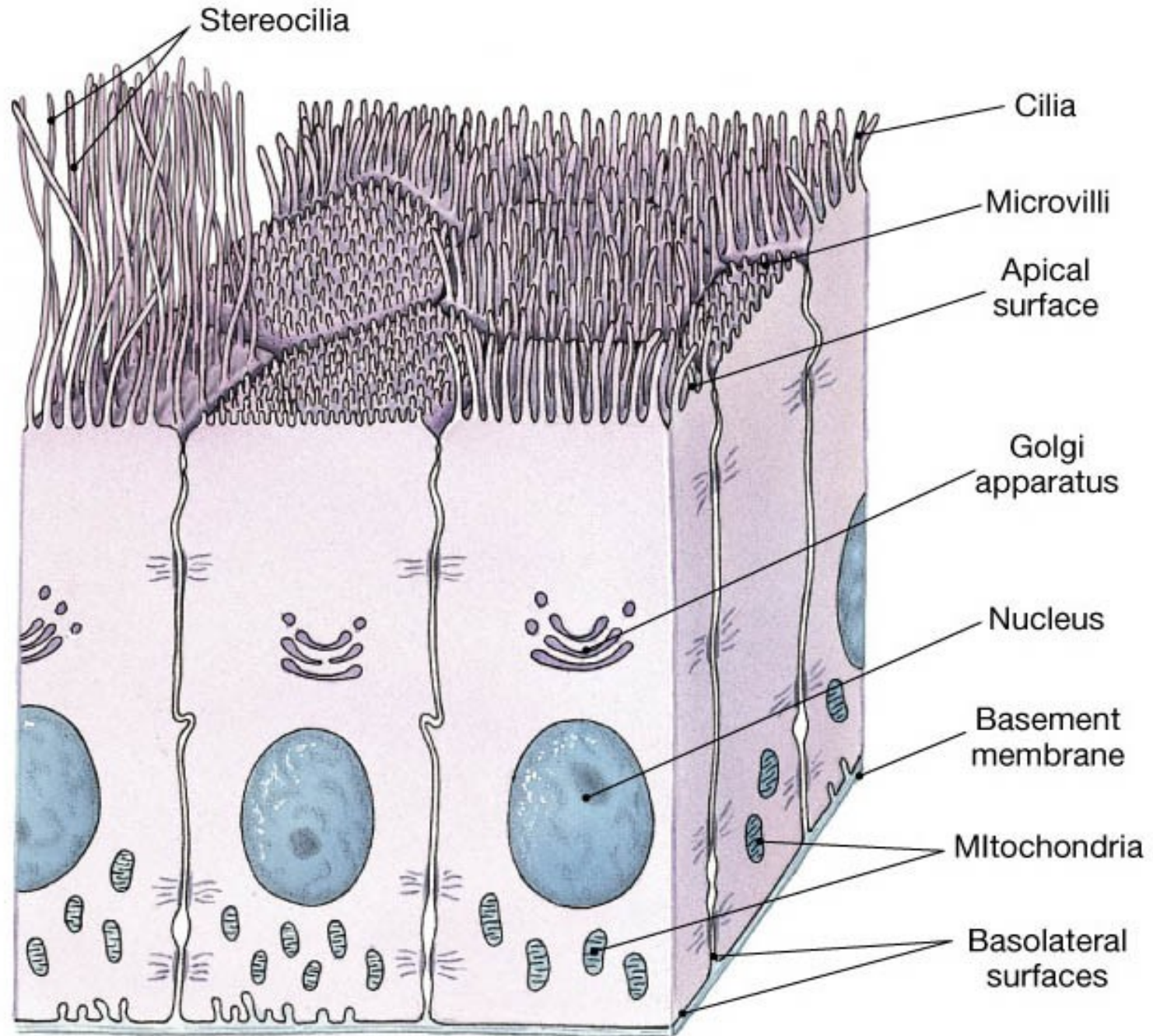
4- Microvilli :

- Are finger like extensions covered cell surface they increase the apical surface area of absorptive cells and called brush border.
- found lining the small intestine and also in proximal



Microvilli- Absorption





(a)

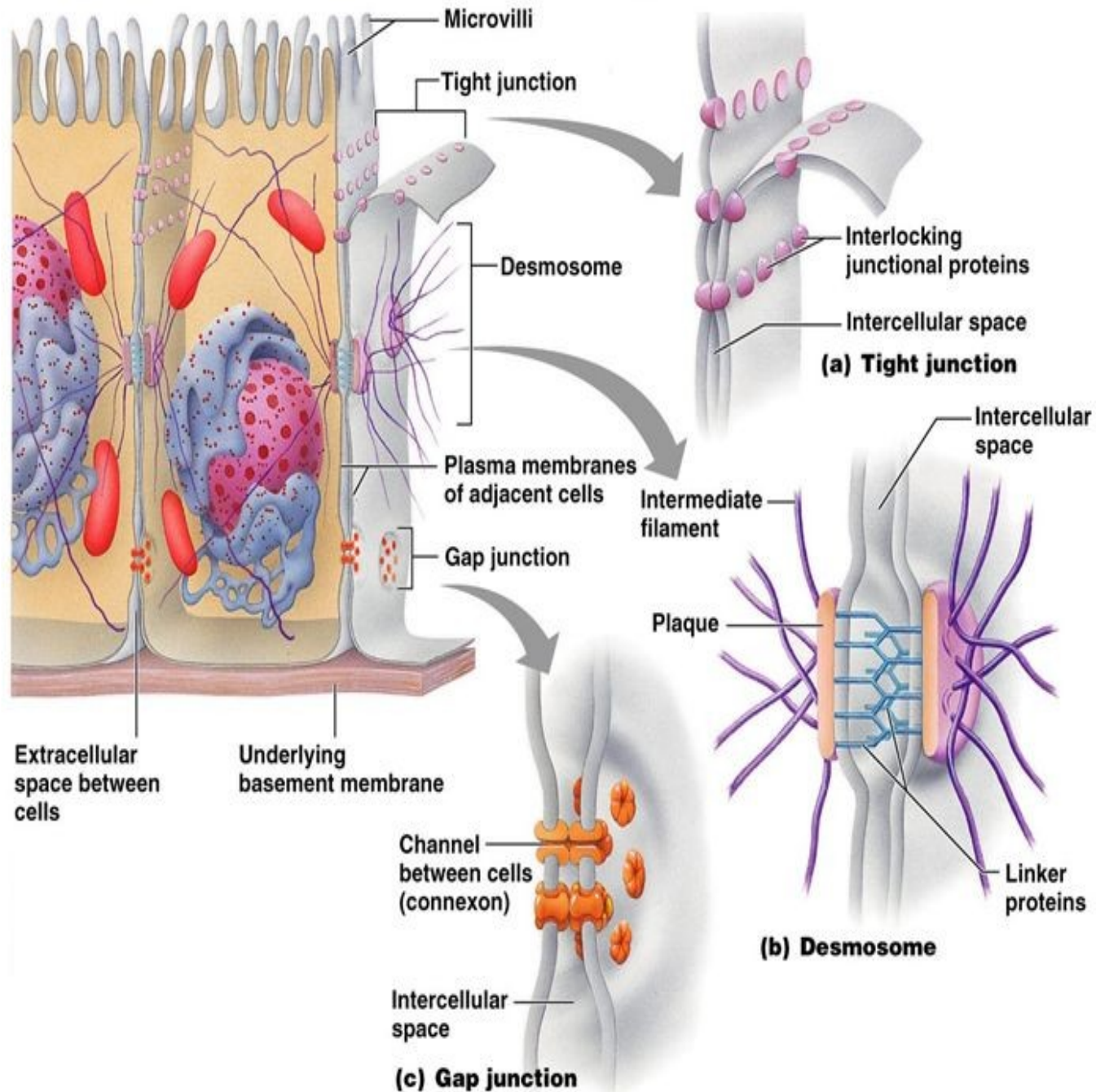
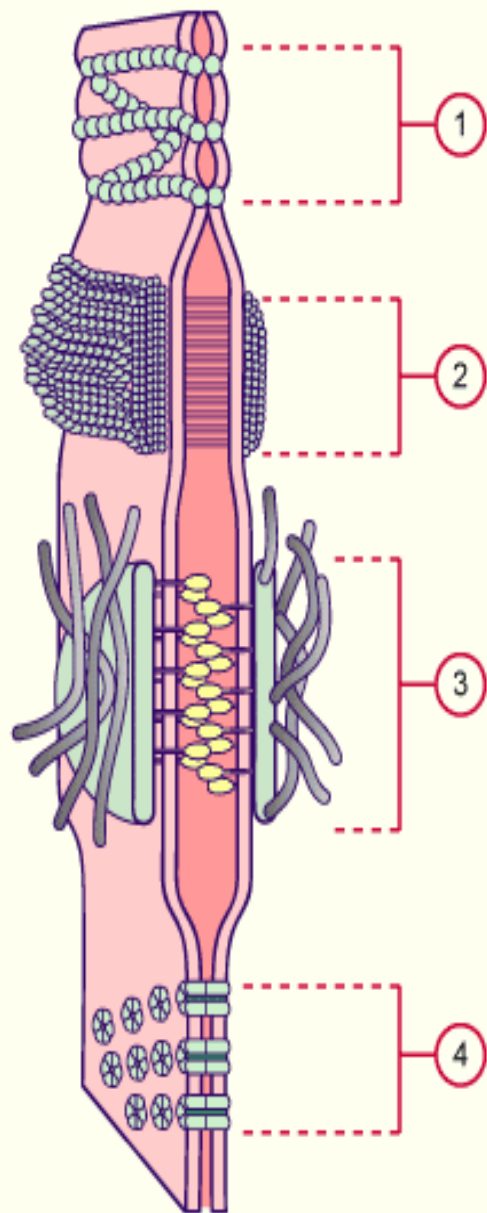
.B- Specialization of the Lateral Surfaces

- The lateral membrane of many epithelial cells often exhibit several types of membrane modifications the intracellular junctions, these junctions from the apex toward the base of the cells are:

- 1-Zonula occludens (tight junctions).
- 2- Zonula adherens (intermediate junctions)
- 3-Desmosome (macula adherens)
- 4- Gap junction

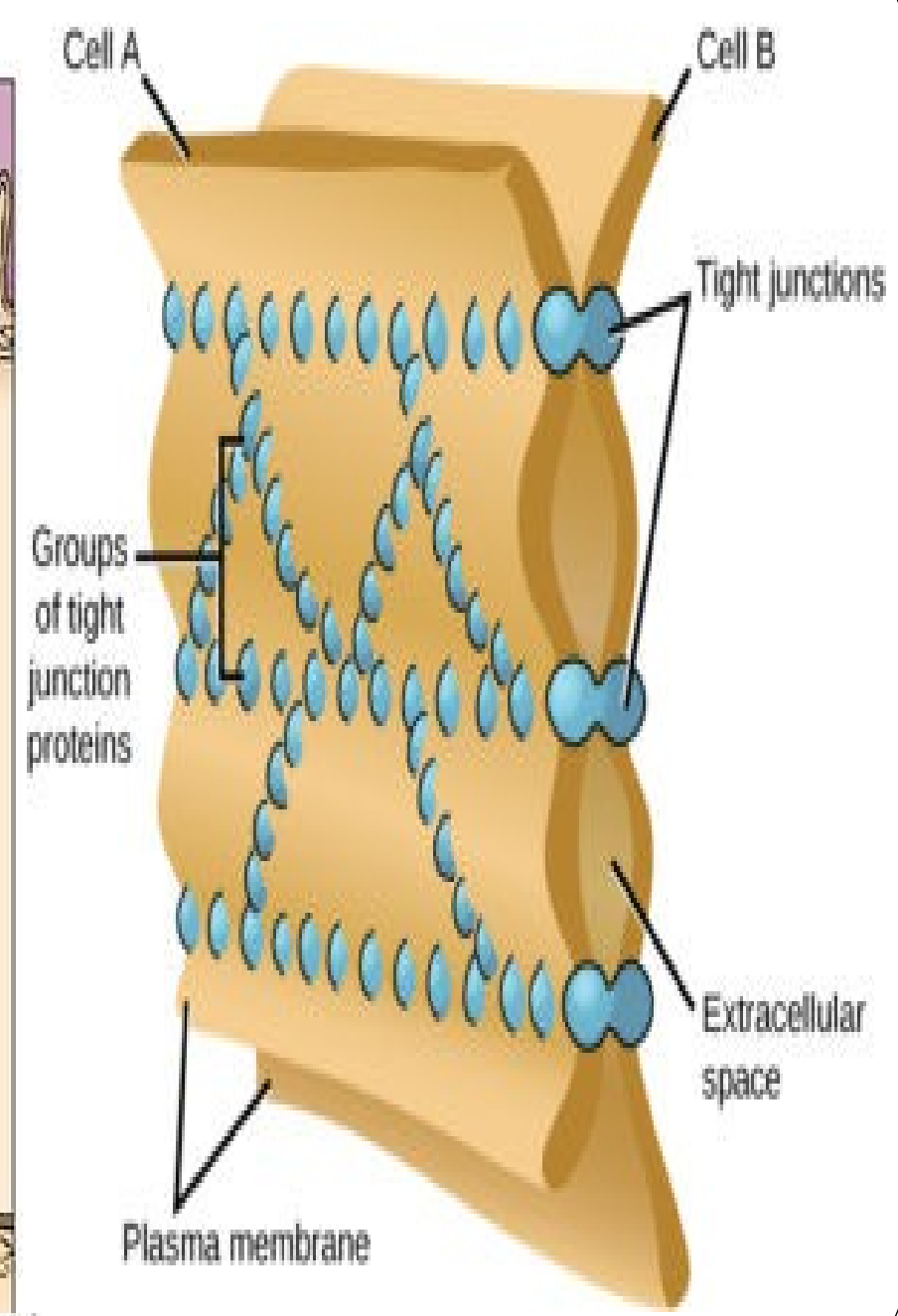
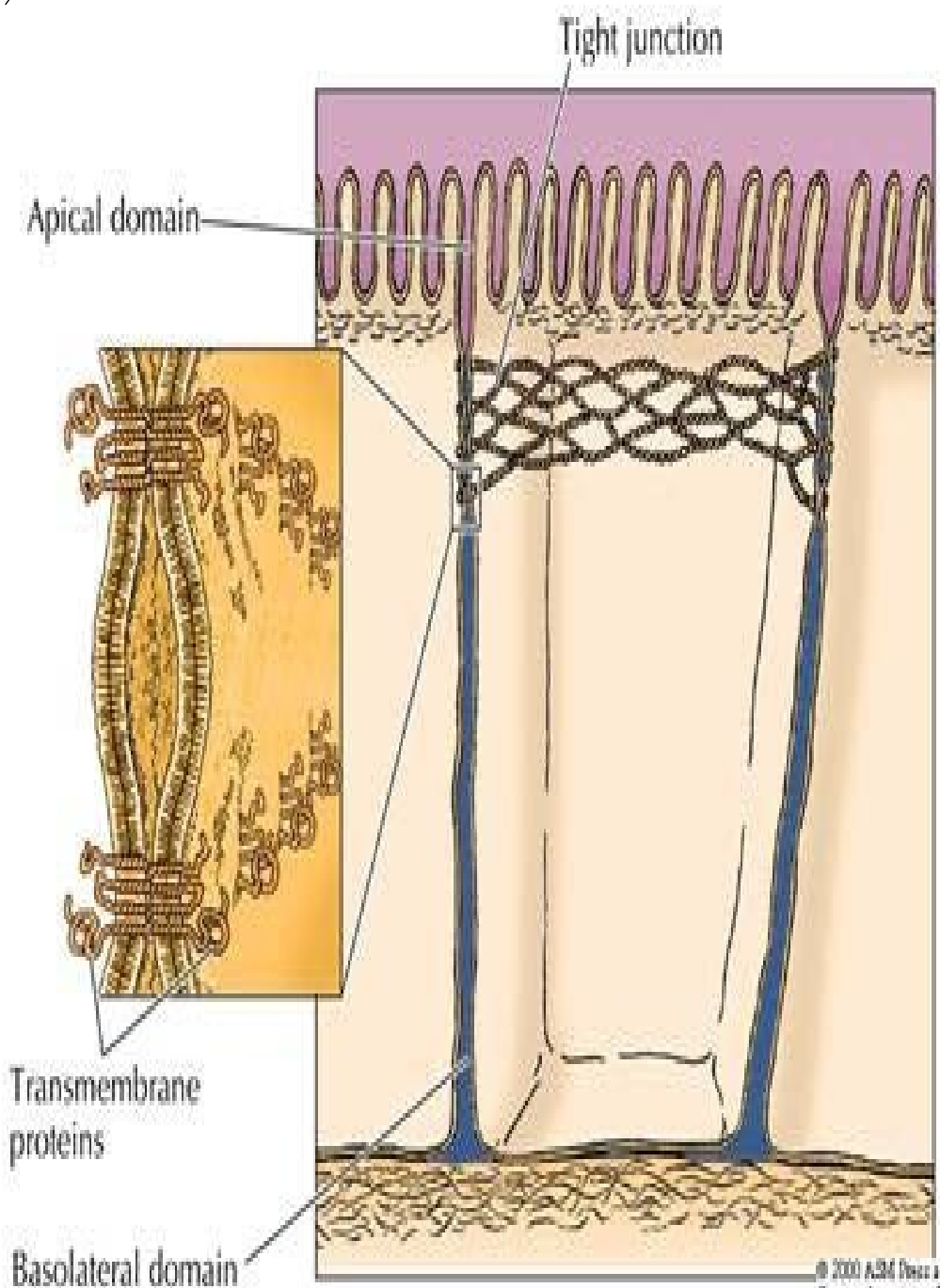
These modification perform three tasks

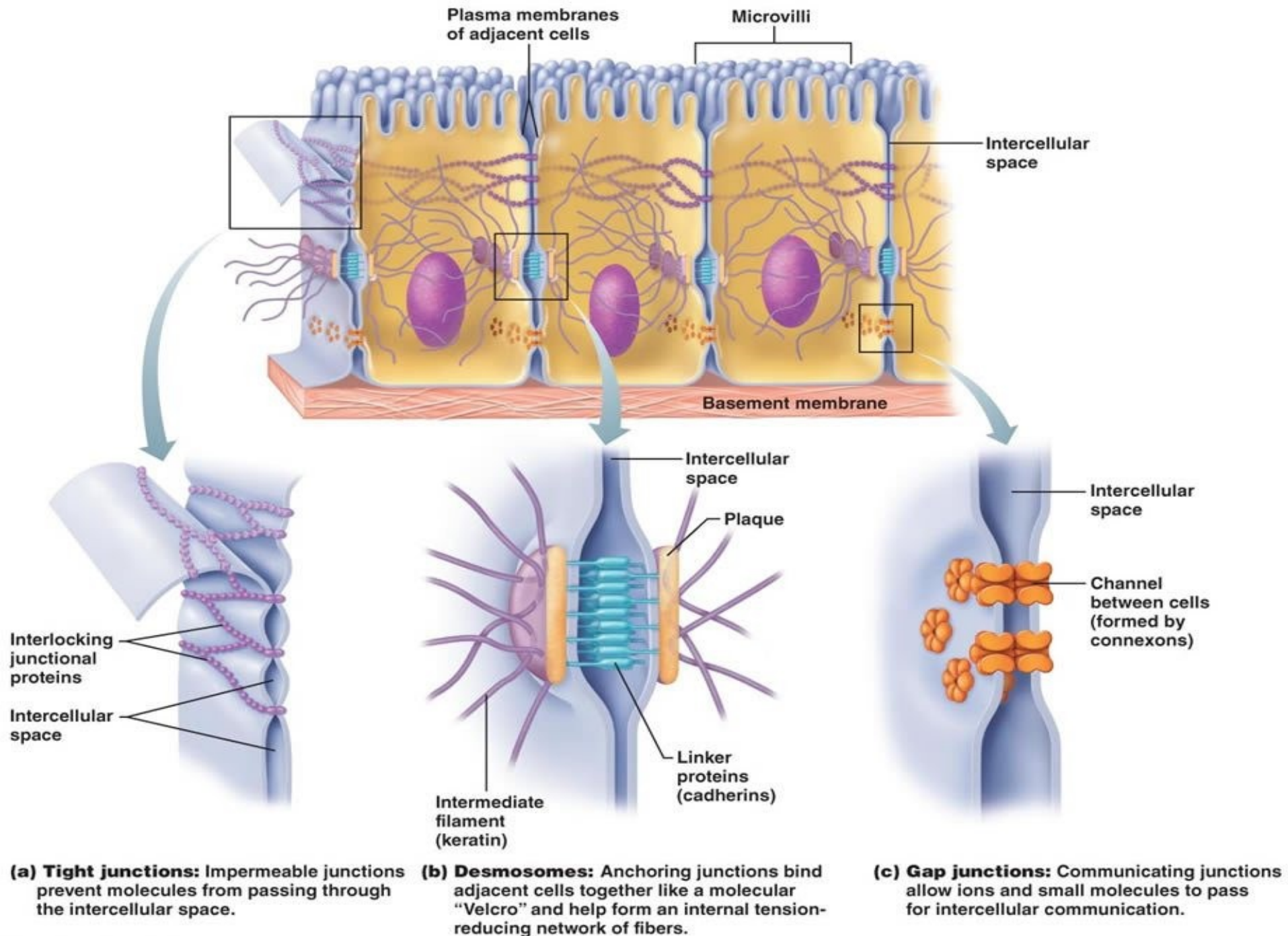
- They mechanically bind the cells together.
- They form permeability barrier.
- They provide a mechanism for intercellular communication



The zonula occludens (ZO) or tight junction (TJ)-1

- Impermeable junctions, the Plasma membranes of neighboring cells are tightly pressed against each other.
 - it act as barrier which Form continuous seals around the cells which prevents leakage of extracellular fluid across a layer of epithelial cells, also prevents passage of molecules and ions through the space between cells.
- ex: they help to keep the intestinal bacteria and toxins out of the blood stream.

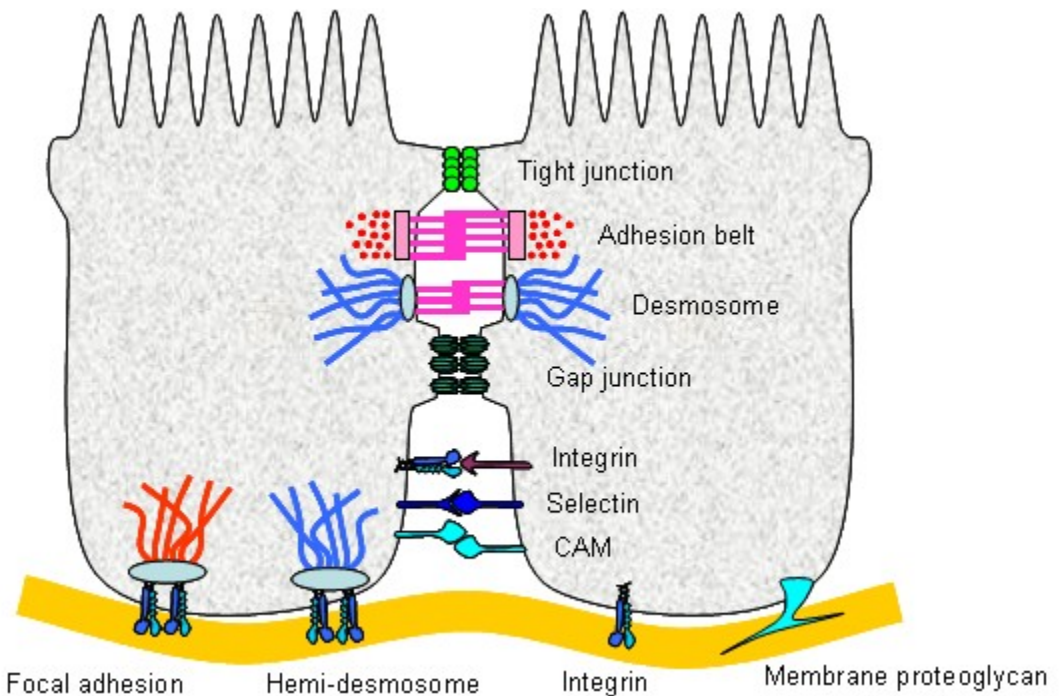


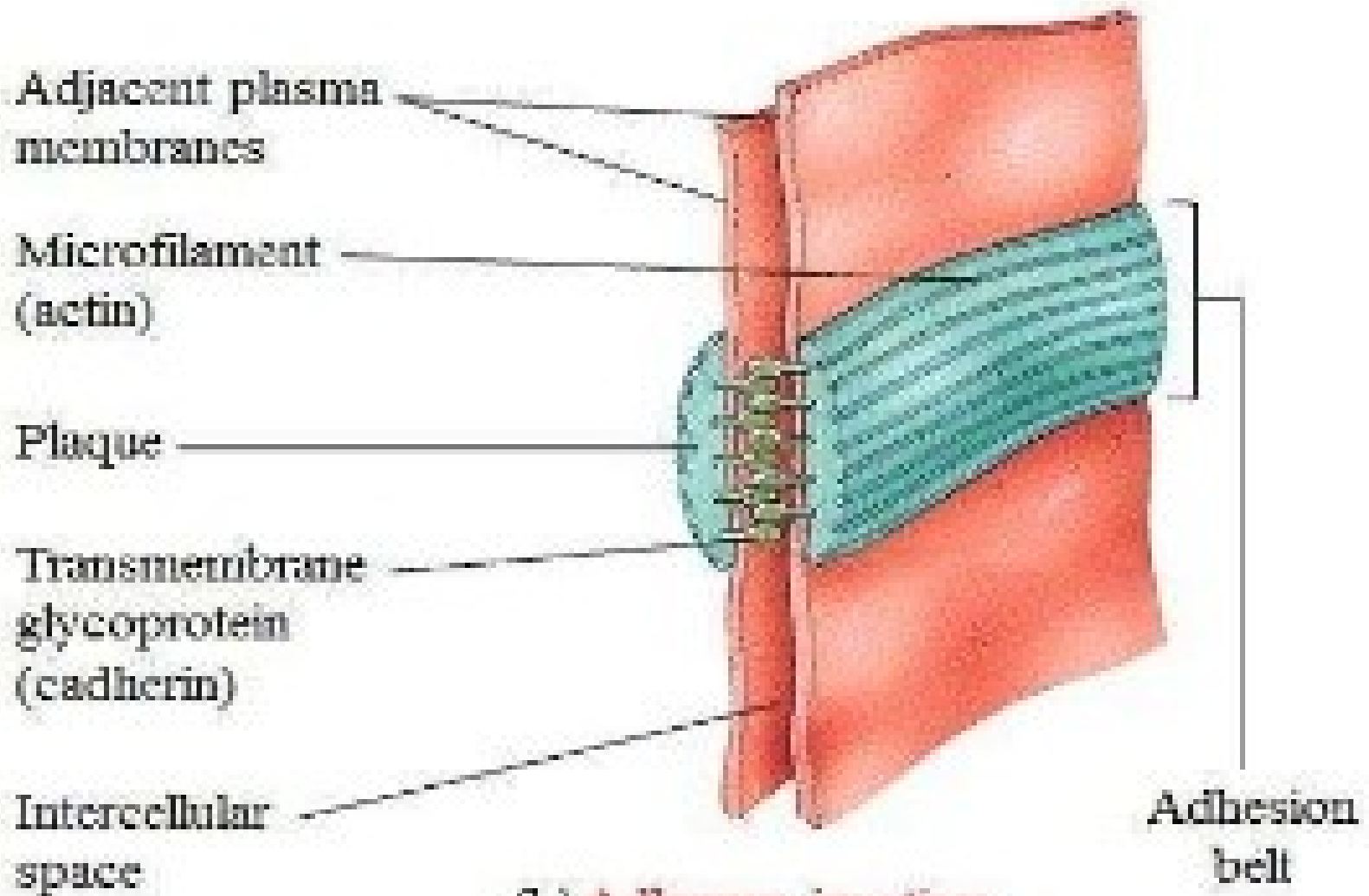


:Zonula adherens=adherens junction -2

This junction encircles the cell and are usually located at basal of the tight junction. composed of 2 electron-dense plaques on opposing cell membranes, containing thin actin micro filaments to strong the joining between adjacent cells.

Joins an actin bundle in one cell to a similar bundle in a neighboring cell.



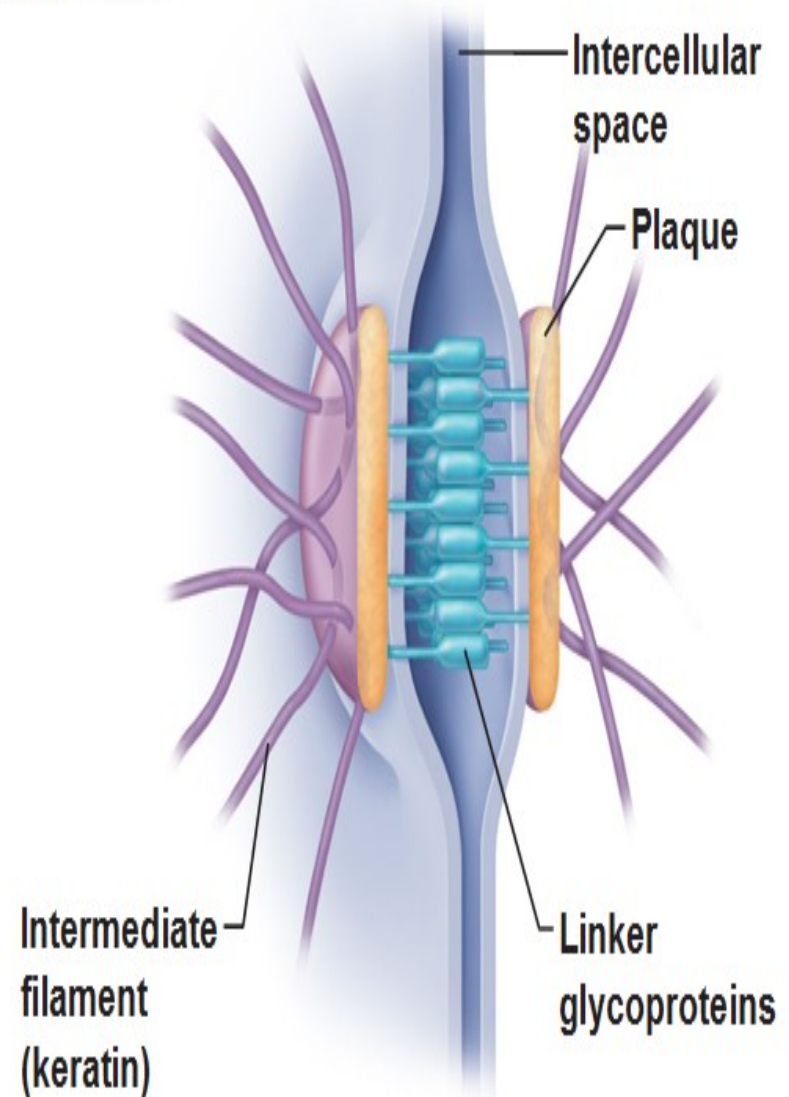


(b) Adherens junction

Desmosomes (Macula adherens) -3

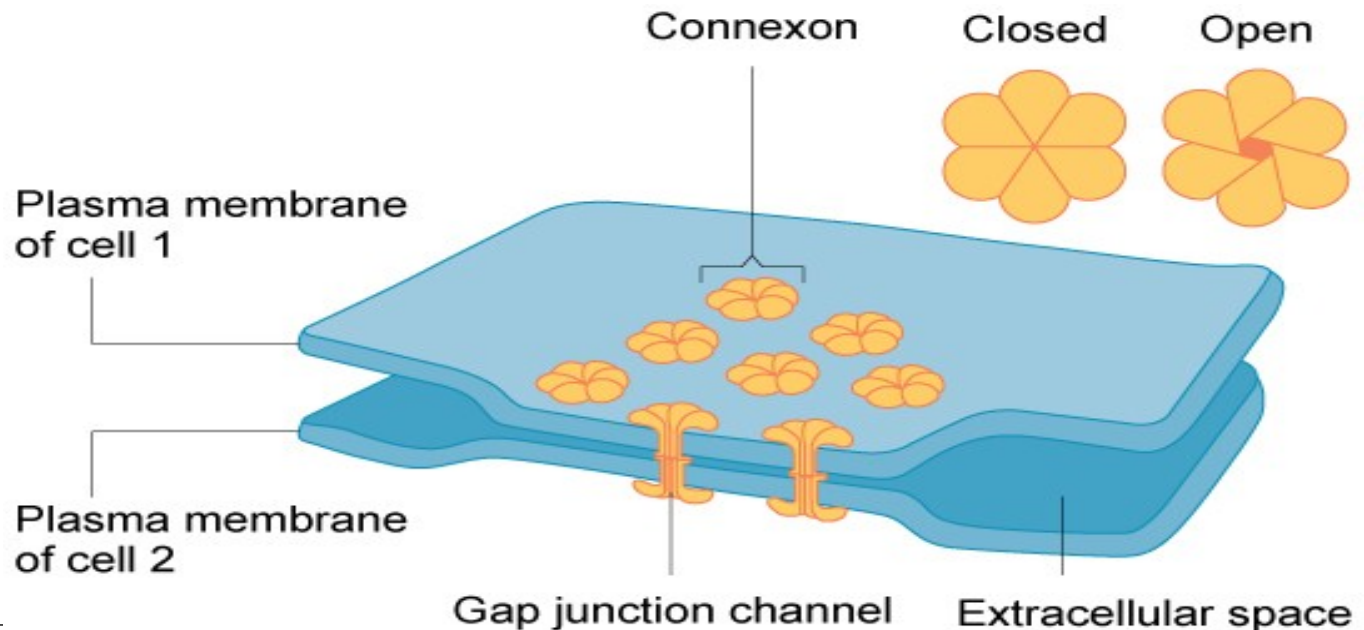
Desmosome

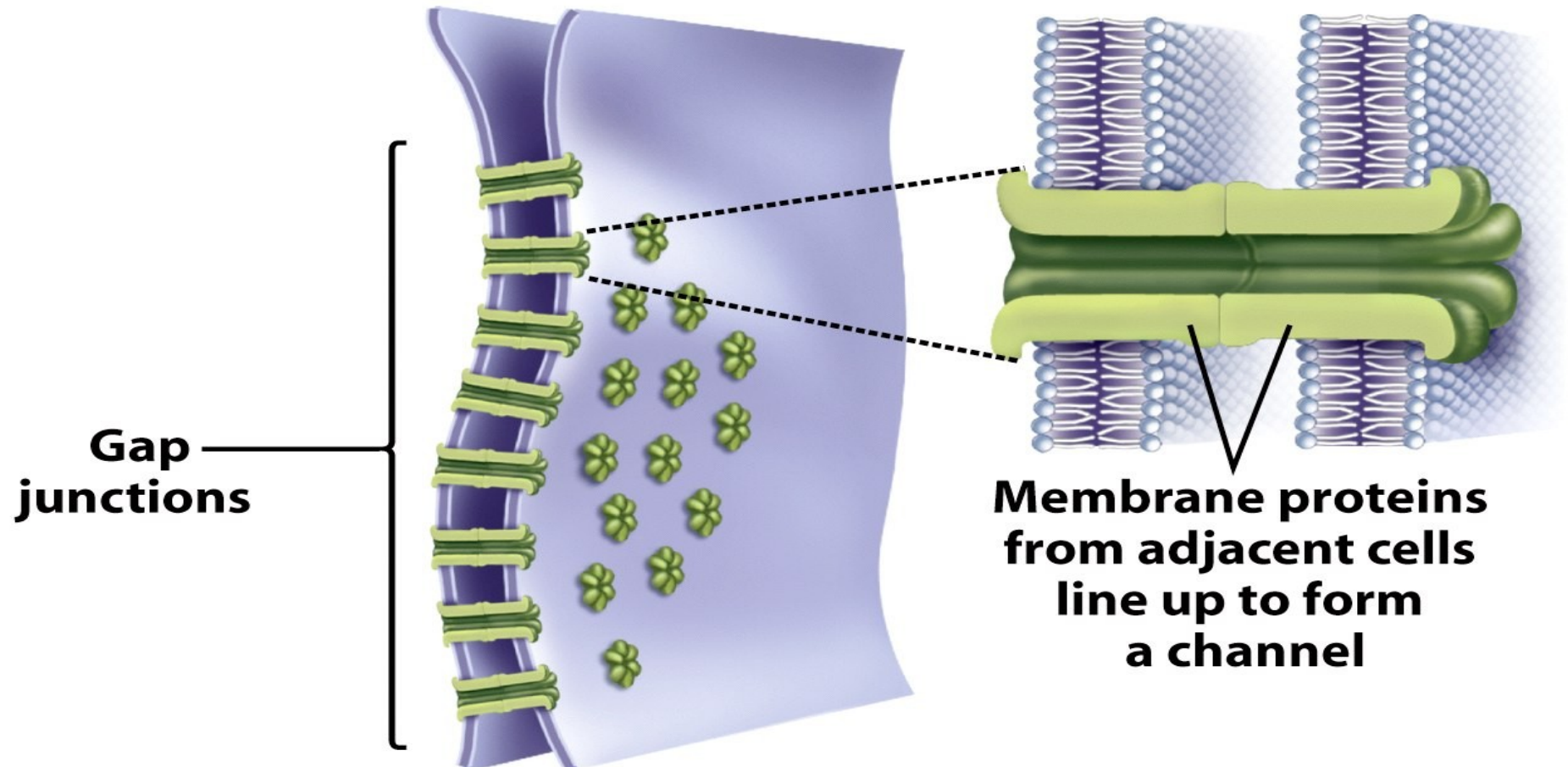
- Is a complex disk-shaped structure (plaque) at the surface of one cell that is match with an identical structure at the surface of the adjacent cell and a group of intermediate filaments are inserted into the attachment plaques ,these filaments are very strong.
- Allow tissues to resist twisting and stretching(stabilize cell shape)



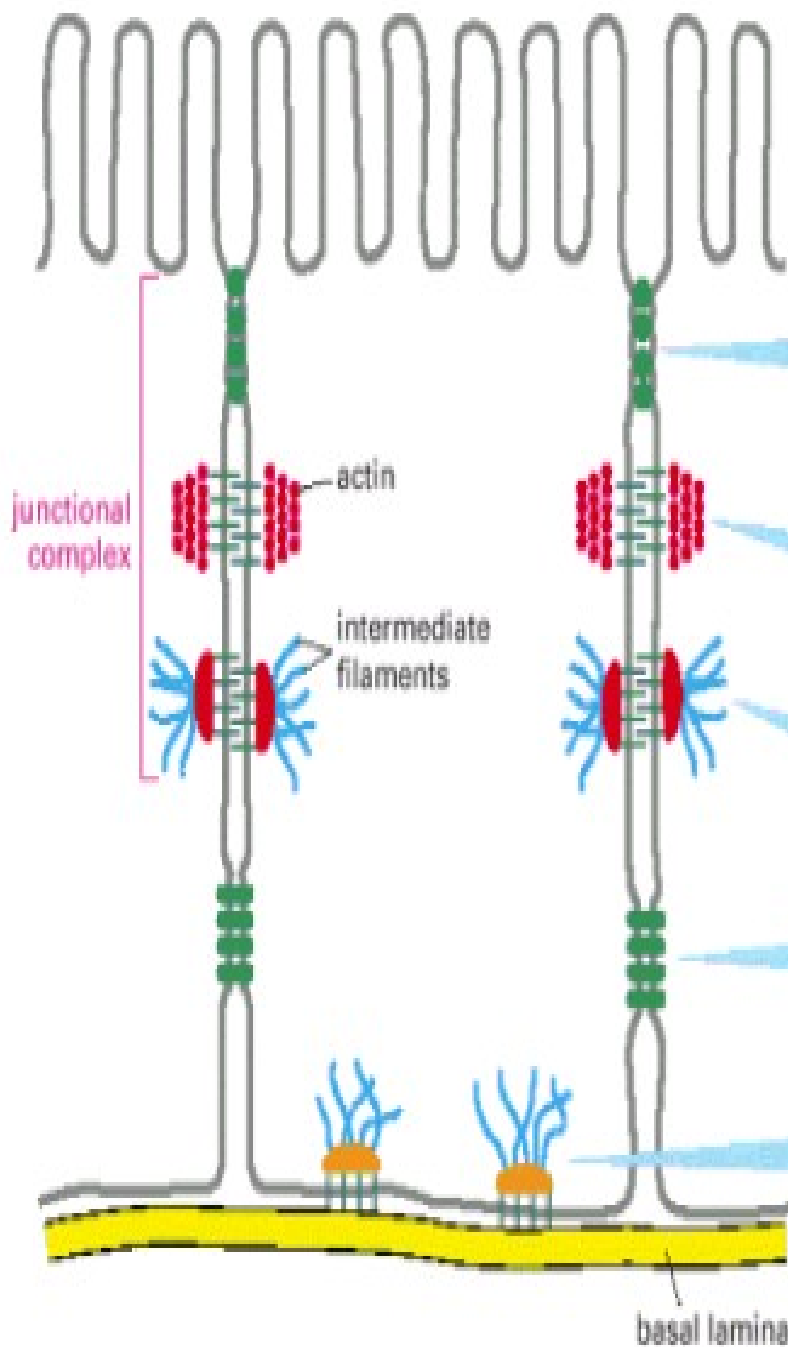
Gap junctions -4

- The individual unit of gap junction is called a **connexon**, which is a membrane proteins form a tunnel between cells.
- Allow cells to communicate with each other, share nutrients, and transfer chemical electrical signals.
- This type is widely distributed in body tissues & best example found in cardiac & smooth muscles.





- Each connexone is formed by 6 gap junction proteins called connexins, which join together leaving a pore form a channels between the two cells, each gap junction composed of tens to hundreds of pairs of connexins.



name	function
tight junction	seals neighboring cells together in an epithelial sheet to prevent leakage of molecules between them
adherens junction	joins an actin bundle in one cell to a similar bundle in a neighboring cell
desmosome	joins the intermediate filaments in one cell to those in a neighbor
gap junction	allows the passage of small water-soluble ions and molecules
hemidesmosome	anchors intermediate filaments in a cell to the basal lamina

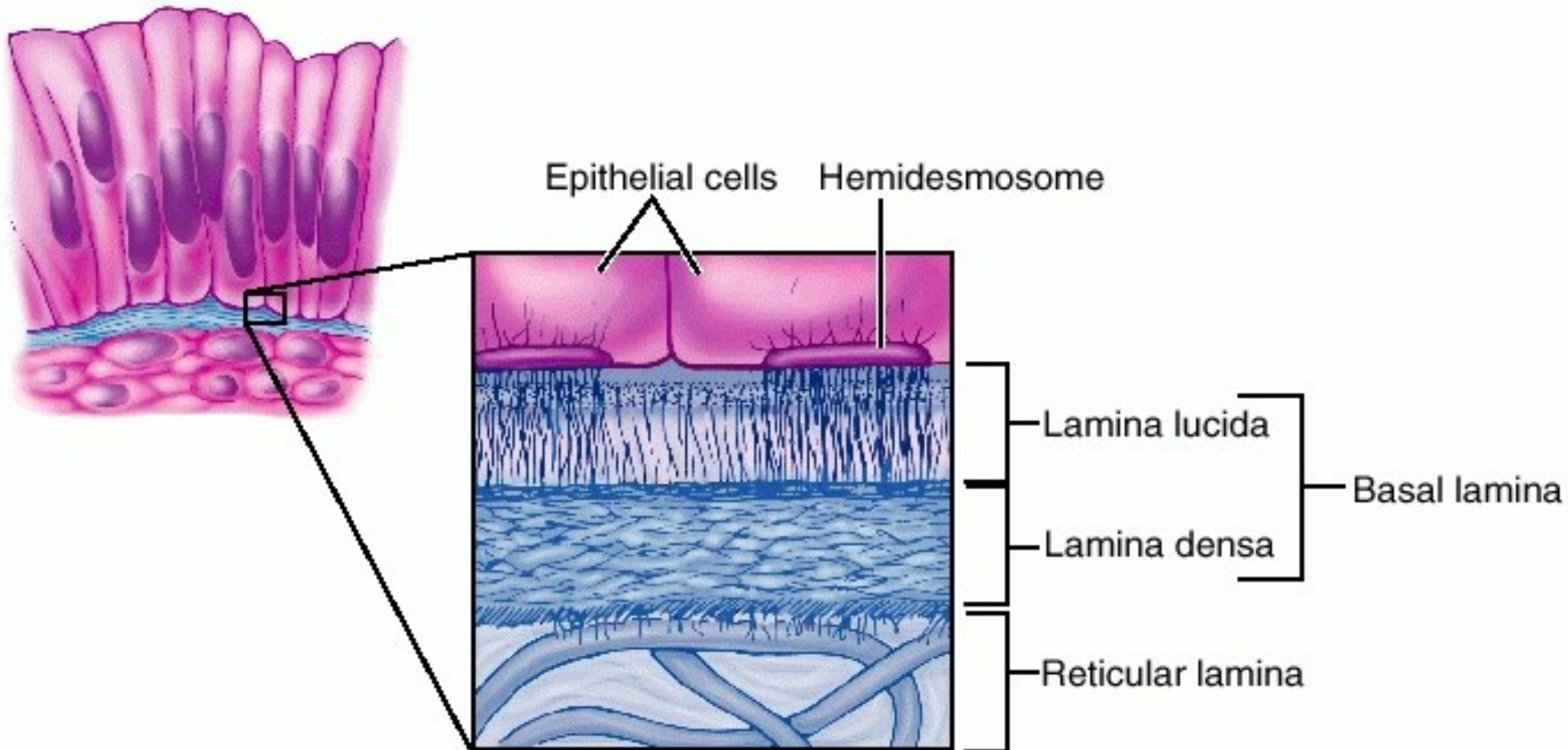
C-Specialization of the Basal Surfaces

1- Basal lamina & Basement membrane

Epithelial cells are separated from the connective tissue by a sheet of extra cellular material called the **basal lamina** ,This structure is visible only with the electron microscope, consisting of :

1. **lamina densa** - a delicate network of very fine fibrils.
 2. **lamina lucida** - electron-lucent layer on one or both sides of the lamina densa.
-
- The functions of basal lamina are:
 1. supporting the cells.
 2. they provide a barrier that limit material exchanges.

- **Basement membrane** is formed by basal lamina and the reticular lamina.
- Reticular lamina is the upper part of the connective tissue .
- Basement membrane is visible with the light microscope.



Basement membrane



2- Hemidesmosome

- These structures take the form of half a desmosome and bind the epithelial cell to the subjacent basement membrane.

